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22/10/2024

1/5/2024

EDA

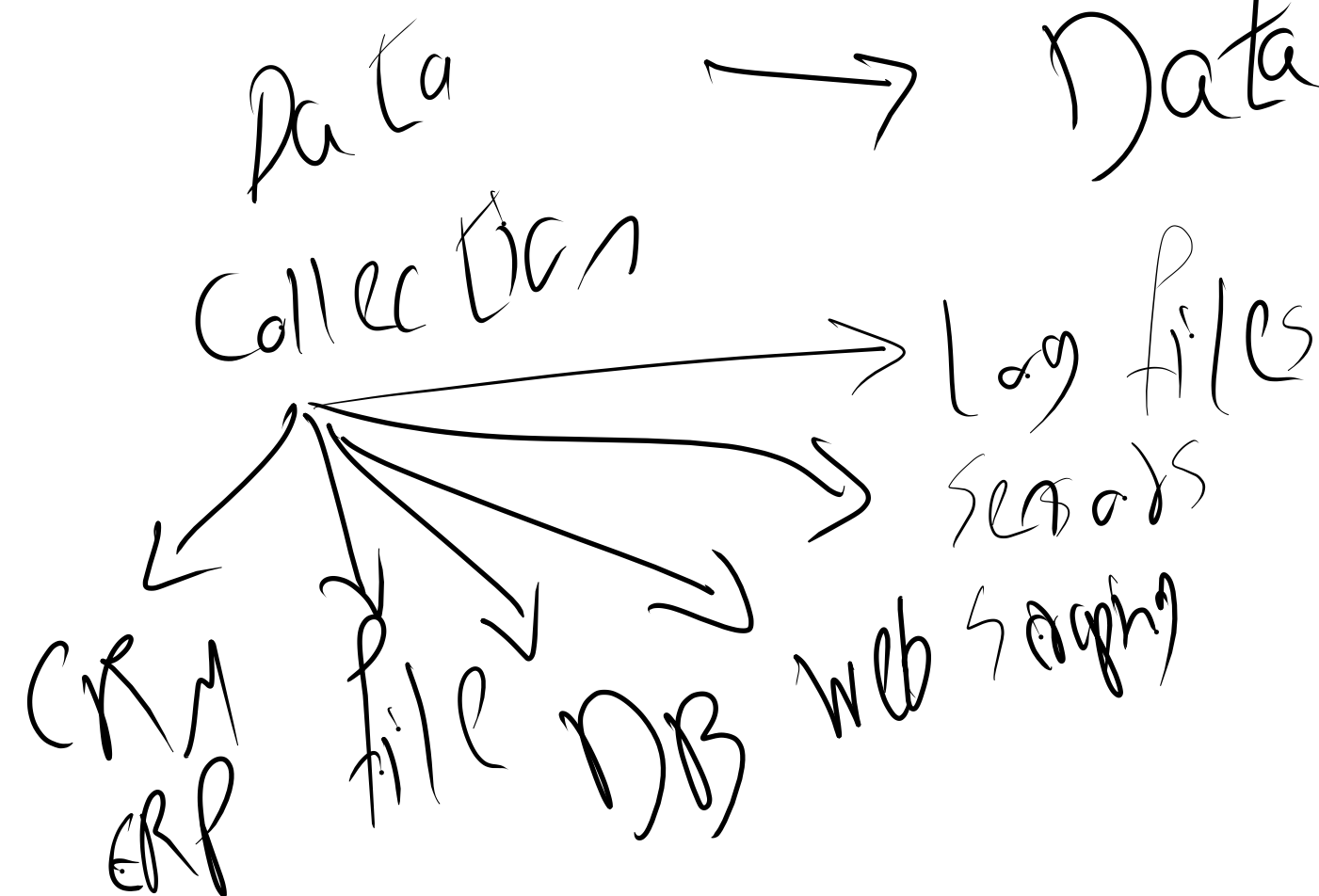
Data science life cycle

(statistics)

Data exploration → Data analysis → Data preprocessing

→ Modeling → Deployment

(Machine Learning)



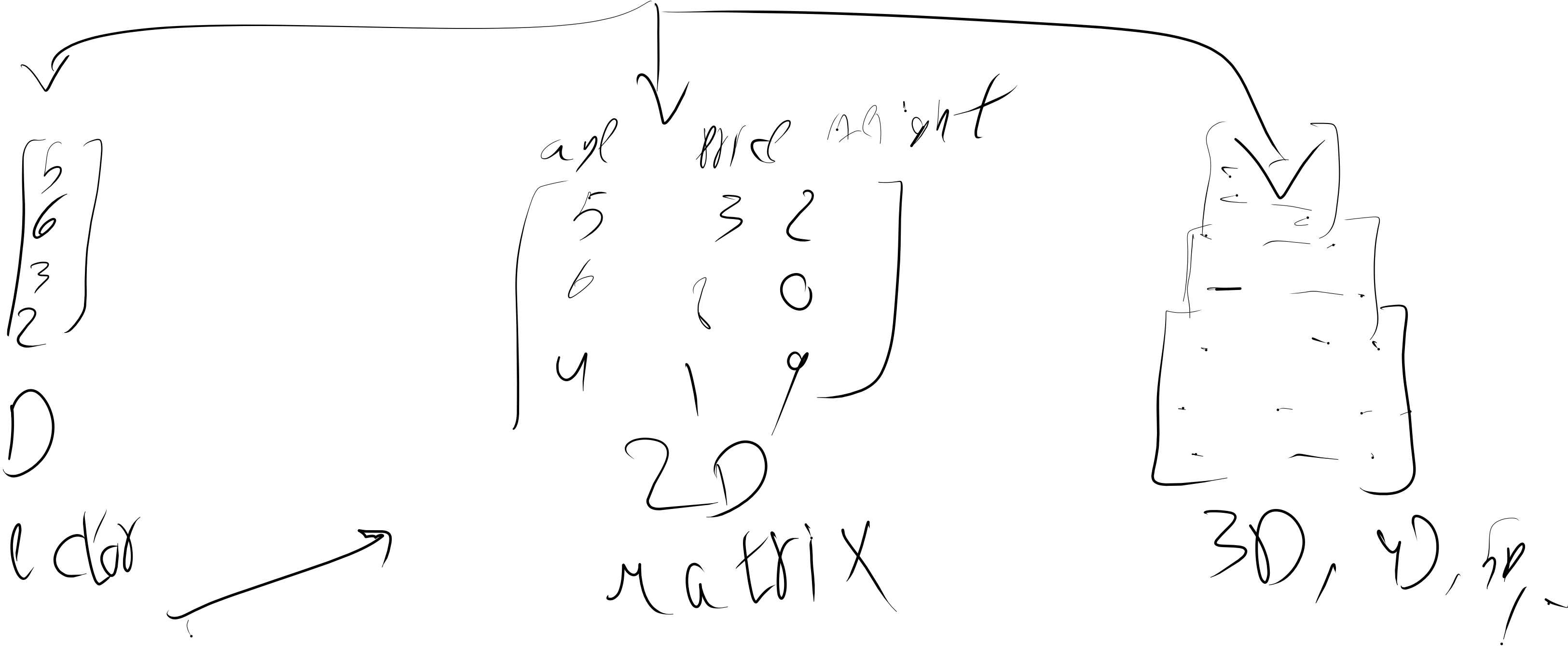
Numpy

Numerical Python

50x faster than python lists

written with C, C++

Array



Dot Product

$$\begin{matrix} \textcircled{1} \rightarrow \\ \textcircled{2} \rightarrow \end{matrix} \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 5 & 6 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 32 \\ 42 \end{bmatrix}$$

Diagram illustrating the dot product calculation for the first row of the resulting array:

Array 1 (2x3): $\begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$

Array 2 (3x2): $\begin{bmatrix} 2 & 4 \\ 5 & 6 \\ 3 & 1 \end{bmatrix}$

Calculation: $(1 \times 2) + (3 \times 5) + (5 \times 3)$

Shape of
Resulting array

$$\begin{bmatrix} 27 \\ 38 \end{bmatrix}$$

2x2